Madison Coots

Harvard University · Cambridge, MA 02138
mcoots@g.harvard.edu | madisoncoots.com

Research Focus

Responsible AI, inequality, discrimination, and the use of computational methods to advance decision-making and policy across a variety of domains, including in healthcare, lending, criminal justice, and education.

Education

Harvard Kennedy School — Cambridge, MA
Ph.D. in Public Policy — *August 2022 - Present*

Harvard University — Cambridge, MA
M.A. in Public Policy — *August 2022 - May 2025*

Stanford University — Stanford, CA
M.S. in Computer Science — *September 2019 - June 2021*
Specialization: *Artificial Intelligence*

Stanford University — Stanford, CA
Bachelor of Science - Management Science and Engineering — *September 2015 - June 2019*
Minor: *English*

Peer-Reviewed Publications

[1] **Automated Court Date Reminders Reduce Warrants for Arrest: Evidence from a Text Messaging Experiment.** Alex Chohlas-Wood, **Madison Coots**, Joe Nudell, Julian Nyarko, Emma Brunskill, Todd Rogers, and Sharad Goel. *Science Advances.* 2025.

[2] **Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates. Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. *Annual Review of Public Health.* 2025.

[3] **A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk. Madison Coots**, Soroush Saghafian, David Kent, and Sharad Goel. *Annals of Internal Medicine.* 2024.

[4] **Learning to be Fair: A Consequentialist Approach to Equitable Decision-Making.** Alex Chohlas-Wood, **Madison Coots**, Henry Zhu, Sharad Goel, and Emma Brunskill. *Management Science.* 2024.

[5] **Designing Equitable Algorithms.** Alex Chohlas-Wood*, **Madison Coots***, Julian Nyarko*, and Sharad Goel*. *Nature Computational Science, Vol. 3. 2023.* *Authors contributed equally.

Research in Progress

[1] **A Profit-Based Measure of Lending Discrimination. Madison Coots**, Robert Bartlett, Julian Nyarko, and Sharad Goel. *Working paper.* 2026.

[2] **Incorporating Text in Empirical Audits of Police Searches. Madison Coots**, Hans Gaebler, Josh Grossman, Alex Chohlas-Wood, Julian Nyarko, and Sharad Goel.

[3] **AI Tutoring and Teacher Support Tools for Elementary Education,** with Calvin Isley, Elisa Xi Chen, Nick Masi, Joe Nudell, Charlotte Tuminelli, Teddy Svoronos, and Sharad Goel.

Honors, Awards, and Fellowships

Dissertation Research Grant, *$10,000* — 2026
- Horowitz Foundation for Social Policy

Rising Star in IOE-ISyE-MS&E — 2026
- Workshop jointly hosted by University of Michigan, Georgia Tech, and Stanford University

Dean's Award for Excellence in Student Teaching — 2025
- Harvard Kennedy School of Government

Distinction in Student Teaching Award 2025

- Harvard Kennedy School of Government

Social Equity and Health Equity Grant, *$5,000* 2025

- Malcolm Weiner Center for Social Policy, Harvard Kennedy School of Government

James M. and Cathleen D. Stone PhD Scholar in Inequality and Wealth Concentration 2023 - Present

- Stone Program in Wealth Distribution, Inequality, and Social Policy, Harvard University

Harvard Graduate Prize Fellowship 2022 - 2023

- Harvard University

Stanford Engineering Coterminal Fellowship 2019 - 2020

- Stanford University

Stokes Scholar 2017 - 2021

- U.S. Federal Government

Selected Presentations

IOE-ISyE-MS&E Rising Stars Workshop Ann Arbor, MI
"A Profit-Based Measure of Lending Discrimination" (Invited talk and poster) *2026*

American Law and Economics Association Conference New York, NY
"A Profit-Based Measure of Lending Discrimination" (Submitted talk) *2025*

Production and Operations Management Society Annual Conference Atlanta, GA
"A Profit-Based Measure of Lending Discrimination" (Invited talk) *2025*

Predictive Analytics & Comparative Effectiveness Center Symposium Boston, MA
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Invited talk) *2024*

Society of Medical Decision Making 46th Annual Meeting Boston, MA
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Submitted talk) *2024*

International Conference on Computational Social Science Philadelphia, PA
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Poster) *2024*

Computational and Methodological Statistics Conference Remote
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Invited talk) *2023*

APPAM Fall Research Conference Atlanta, GA
"Automated Court Date Reminders Reduce Warrants for Arrest" (Submitted talk) *2023*

INFORMS General Meeting Phoenix, AZ
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Invited talk) *2023*

INFORMS Healthcare Conference Toronto, ON
"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk" (Invited talk) *2023*

ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization Arlington, VA
"Learning to be Fair" (Submitted talk with Alex Chohlas-Wood) *2022*

Teaching

Harvard Kennedy School of Government

- **Teaching Fellow.** API 201: Quantitative Analysis and Empirical Methods I (Fall 2024, 2025). *Graduate course in applied statistics: exploring and summarizing data with R, probability theory, statistical inference, decision analysis.*
- **Teaching Fellow.** API 202Z: Quantitative Analysis and Empirical Methods II (Spring 2025, 2026). *Advanced graduate course in applied statistics: statistical inference and linear regression models.*
- **Teaching Fellow.** API 203Z: Quantitative Analysis and Empirical Methods III (Spring 2025). *Advanced graduate course in applied statistics: causal inference and machine learning.*
- **Teaching Fellow.** DPI 681M: The Science and Implications of Generative AI (Spring 2024) *Designed and taught the technical complement to the course on language models, deep learning models, and transformers.*

Stanford University, School of Engineering

- **Course Assistant.** MS&E 125: Applied Statistics (Winter 2020) *Undergraduate course in applied statistics: exploring and summarizing data, statistical inference, linear and logistic regression models.*
- **Course Assistant.** MS&E 252: Foundations of Decision Analysis (Fall 2019). *Graduate course in quantitative decision analysis covering: utility theory, sensitivity analysis, value of information. Recognized by Stanford Center for Professional Development for excellence in teaching.*

Stanford Code in Place Program

- **Section Leader.** (Spring 2020). *Part of a teaching team for Code in Place, offered by Stanford during COVID-19 pandemic, with 10,000 global students and 900 volunteer teachers participating from around the world. Prepared and taught a weekly discussion section of 10-12 students to supplement professors' lectures in a 5-week introductory online Python programming course.*

Professional Service

Peer Review

Ad Hoc Reviewer

- *Epidemiology* (2025)
- *International Conference on Computational Social Science* (2025, 2026)

Skills Summary

Languages: R, Python

AI: LLM prompt and response evaluation, text data processing and information extraction, chatbot design and user experience for AI systems

Methods: Stochastic modeling, decision analysis, statistical inference, causal inference

Professional Experience

Senior Data Scientist (Part-time) — Woburn, MA
Systems & Technology Research — *Feb. 2023 - Present*

Data Scientist — Boston, MA
Stanford Computational Policy Lab — *Sep. 2020 - Aug. 2022*

Data Scientist (Part-time) — Remote
Aerospace Technical Services — *Sep. 2020 - Jun. 2024*

Data Science Intern — Washington D.C.
U.S. Federal Government (agency withheld) — *Jun. 2017 - Jan. 2021*